

CARL MUGI NJEKE

+254 790 457116 | njekecarl@gmail.com | Nairobi, Kenya | www.linkedin.com/in/carl-njeke-9a753422b

PROFILE

Biomedical Engineering graduate from the University of Bradford with a strong interest in biomedical science, healthcare technology, laboratory systems, and healthcare innovation. Experienced in clinical operations support, medical documentation, and user-centered healthcare innovation through the IVIE CDIE program. Passionate about applying scientific knowledge, research principles, and emerging technologies to improve healthcare delivery. Recognized for strong ethical standards, analytical thinking, teamwork, and commitment to continuous learning.

EDUCATION

BSc Biomedical Engineering

University of Bradford, UK | 2021 - 2024

Relevant coursework:

- Embedded Electronics
- Healthcare Technology Project
- Computer-Aided Engineering
- Clinical Movement Analysis
- Functional Anatomy and Human Physiological Measurements
- Biomaterials with Implant Design and Technology
- Medical Ethics and Quality Management
- Rehabilitation and Prosthetics

IB Diploma

Aga Khan Academy, Mombasa | 2019 - 2021

- Total score: 36/45; Higher Level: Economics (6), Biology (6), Chemistry (5); Standard Level: Math A&A (6), Eng & Literature (6), Swahili (5).

IGCSE Certificate

Nairobi Academy | 2017 - 2019

- Biology (A), Chemistry (A), Physics (A), Mathematics (B), Business (C), Geography (B), ICT (B), English (A).

CERTIFICATION

Biomedical Equipment: Repairing and Maintaining Biomedical Devices

Delft University of Technology (TU Delft) - edX Online Learning | 2026

- Medical device classification and functionality
- Preventive maintenance planning
- Structured troubleshooting
- Biomedical safety and risk management
- Equipment lifecycle workflow

EXPERIENCE

Clinical Operations Support

Convergent Medical Services, Nairobi | Jan 2025 - Present

- Supported nurses in conducting occupational medical examinations for employees across Kenya, including physical assessments, audiometry, spirometry, laboratory tests, and clinical documentation.
- Generated, managed, and updated Certificates of Fitness through the DOSHS platform while maintaining accurate employee medical records.
- Ensured completeness, accuracy, and compliance of medical documentation in line with occupational health, safety, and healthcare standards.
- Improved clinical workflow efficiency by coordinating examination records, reducing turnaround time, and supporting smooth patient and employer service delivery.

Healthcare Innovation Participant - IVIE CDIE Summer Program

Kenyatta University | May 2026

- Completed an intensive healthcare innovation program focused on need identification, user-centered design, prototyping, budgeting, and pitching.
- Worked on Neo Care, a neonatal oxygen monitoring concept inspired by observations in overcrowded, low-resource neonatal units.
- Developed a solution concept to help nurses monitor multiple neonates from a central dashboard and receive alerts when oxygen saturation becomes abnormal.
- Gained practical experience translating clinical observations into a viable need statement, concept note, prototype direction, and implementation budget.
- Collaborated within a multidisciplinary team to evaluate feasibility, prototype concepts, and healthcare implementation strategies.

PROJECTS

Neo Care - Neonatal Oxygen Monitoring Concept

IVIE CDIE, Kenyatta University | 2026

- Identified a clinical need to improve continuous oxygen saturation monitoring for neonates in overcrowded low-resource neonatal units.
- Proposed a multi-patient monitoring approach designed to reduce delayed detection of hypoxia and support overstretched nursing teams.
- Contributed to concept development, prototype planning, budget preparation, and pitch communication.

Wearable Assessment of Angular Trunk Motion in Chronic Back Pain

University of Bradford | 2024

- Investigated the application of wearable sensor technology for monitoring trunk motion in individuals with chronic back pain.
 - Collected, analysed, and interpreted movement data to support evidence-based assessment of biomechanical function.
 - Applied scientific research methods, data analysis techniques, and clinical movement analysis principles throughout the project.
 - Produced technical documentation, presented findings and possible further advancements.
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KEY SKILLS & COMPETENCIES

Technical Skills: Medical equipment support, healthcare technologies, embedded electronics fundamentals, basic troubleshooting, AutoCAD, SolidWorks, 3D design, Microsoft Office, digital record management, technical documentation.

Clinical & Professional Skills: Clinical operations support, medical documentation, compliance awareness, need identification, concept development, prototyping, budgeting, pitching, teamwork, leadership, communication, adaptability, and cross-cultural collaboration.